

## ML-LAB PRACTICAL-01

```
import pandas as pd
```

Choose Files Iris.csv.xls  
Iris.csv.xls(application/vnd.ms-excel) - 5107 bytes, last modified: 3/4/2026 - 100% done  
Saving Iris.csv.xls to Iris.csv (3).xls

```
df=pd.read_csv("/content/Iris.csv.xls")
print(df)
```

```
[150 rows x 6 columns]
```

```
Column Name: Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
                    'Species'],
                    dtype='object')
Target Classes: ['Iris-setosa' 'Iris-versicolor' 'Iris-virginica']
```

```
df.head()
```

|   | Id | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | Species     |  |
|---|----|---------------|--------------|---------------|--------------|-------------|--|
| 0 | 1  | 5.1           | 3.5          | 1.4           | 0.2          | Iris-setosa |  |
| 1 | 2  | 4.9           | 3.0          | 1.4           | 0.2          | Iris-setosa |  |
| 2 | 3  | 4.7           | 3.2          | 1.3           | 0.2          | Iris-setosa |  |
| 3 | 4  | 4.6           | 3.1          | 1.5           | 0.2          | Iris-setosa |  |
| 4 | 5  | 5.0           | 3.6          | 1.4           | 0.2          | Iris-setosa |  |

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

df.tail()

|     | Id  | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | Species        |  |
|-----|-----|---------------|--------------|---------------|--------------|----------------|--|
| 145 | 146 | 6.7           | 3.0          | 5.2           | 2.3          | Iris-virginica |  |
| 146 | 147 | 6.3           | 2.5          | 5.0           | 1.9          | Iris-virginica |  |
| 147 | 148 | 6.5           | 3.0          | 5.2           | 2.0          | Iris-virginica |  |
| 148 | 149 | 6.2           | 3.4          | 5.4           | 2.3          | Iris-virginica |  |
| 149 | 150 | 5.9           | 3.0          | 5.1           | 1.8          | Iris-virginica |  |

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column              Non-Null Count  Dtype
---  -
0    Id                   150 non-null   int64
1    SepalLengthCm       150 non-null   float64
2    SepalWidthCm        150 non-null   float64
3    PetalLengthCm       150 non-null   float64
4    PetalWidthCm        150 non-null   float64
5    Species              150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

df.describe()

|       | Id         | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm |  |
|-------|------------|---------------|--------------|---------------|--------------|--|
| count | 150.000000 | 150.000000    | 150.000000   | 150.000000    | 150.000000   |  |
| mean  | 75.500000  | 5.843333      | 3.054000     | 3.758667      | 1.198667     |  |
| std   | 43.445368  | 0.828066      | 0.433594     | 1.764420      | 0.763161     |  |
| min   | 1.000000   | 4.300000      | 2.000000     | 1.000000      | 0.100000     |  |
| 25%   | 38.250000  | 5.100000      | 2.800000     | 1.600000      | 0.300000     |  |
| 50%   | 75.500000  | 5.800000      | 3.000000     | 4.350000      | 1.300000     |  |
| 75%   | 112.750000 | 6.400000      | 3.300000     | 5.100000      | 1.800000     |  |
| max   | 150.000000 | 7.900000      | 4.400000     | 6.900000      | 2.500000     |  |

print('Number of rows and columns:',df.shape)

Number of rows and columns: (150, 6)

df.dropna(inplace=True)  
print(df)

|     | Id  | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | \ |
|-----|-----|---------------|--------------|---------------|--------------|---|
| 0   | 1   | 5.1           | 3.5          | 1.4           | 0.2          |   |
| 1   | 2   | 4.9           | 3.0          | 1.4           | 0.2          |   |
| 2   | 3   | 4.7           | 3.2          | 1.3           | 0.2          |   |
| 3   | 4   | 4.6           | 3.1          | 1.5           | 0.2          |   |
| 4   | 5   | 5.0           | 3.6          | 1.4           | 0.2          |   |
| ..  | ... | ...           | ...          | ...           | ...          |   |
| 145 | 146 | 6.7           | 3.0          | 5.2           | 2.3          |   |

```

146 147      6.3      2.5      5.0      1.9
147 148      6.5      3.0      5.2      2.0
148 149      6.2      3.4      5.4      2.3
149 150      5.9      3.0      5.1      1.8

```

```

      Species
0      Iris-setosa
1      Iris-setosa
2      Iris-setosa
3      Iris-setosa
4      Iris-setosa
..      ...
145 Iris-virginica
146 Iris-virginica
147 Iris-virginica
148 Iris-virginica
149 Iris-virginica

```

```
[150 rows x 6 columns]
```

```
print('Number of null values:',df.isnull().sum())
```

```

Number of null values: Id      0
SepalLengthCm      0
SepalWidthCm      0
PetalLengthCm      0
PetalWidthCm      0
Species      0
dtype: int64

```

Compute statistical measures such as mean,median,std deviation,min,max using describe()

```
df.describe()
```

|              | Id         | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm |
|--------------|------------|---------------|--------------|---------------|--------------|
| <b>count</b> | 150.000000 | 150.000000    | 150.000000   | 150.000000    | 150.000000   |
| <b>mean</b>  | 75.500000  | 5.843333      | 3.054000     | 3.758667      | 1.198667     |
| <b>std</b>   | 43.445368  | 0.828066      | 0.433594     | 1.764420      | 0.763161     |
| <b>min</b>   | 1.000000   | 4.300000      | 2.000000     | 1.000000      | 0.100000     |
| <b>25%</b>   | 38.250000  | 5.100000      | 2.800000     | 1.600000      | 0.300000     |
| <b>50%</b>   | 75.500000  | 5.800000      | 3.000000     | 4.350000      | 1.300000     |
| <b>75%</b>   | 112.750000 | 6.400000      | 3.300000     | 5.100000      | 1.800000     |
| <b>max</b>   | 150.000000 | 7.900000      | 4.400000     | 6.900000      | 2.500000     |

Observe shape and sample records

```
df.shape
```

```
(150, 6)
```

```
df.sample(5)
```

|            | Id  | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | Species         |
|------------|-----|---------------|--------------|---------------|--------------|-----------------|
| <b>6</b>   | 7   | 4.6           | 3.4          | 1.4           | 0.3          | Iris-setosa     |
| <b>12</b>  | 13  | 4.8           | 3.0          | 1.4           | 0.1          | Iris-setosa     |
| <b>8</b>   | 9   | 4.4           | 2.9          | 1.4           | 0.2          | Iris-setosa     |
| <b>134</b> | 135 | 6.1           | 2.6          | 5.6           | 1.4          | Iris-virginica  |
| <b>85</b>  | 86  | 6.0           | 3.4          | 4.5           | 1.6          | Iris-versicolor |

Analyze the number of samples in each class

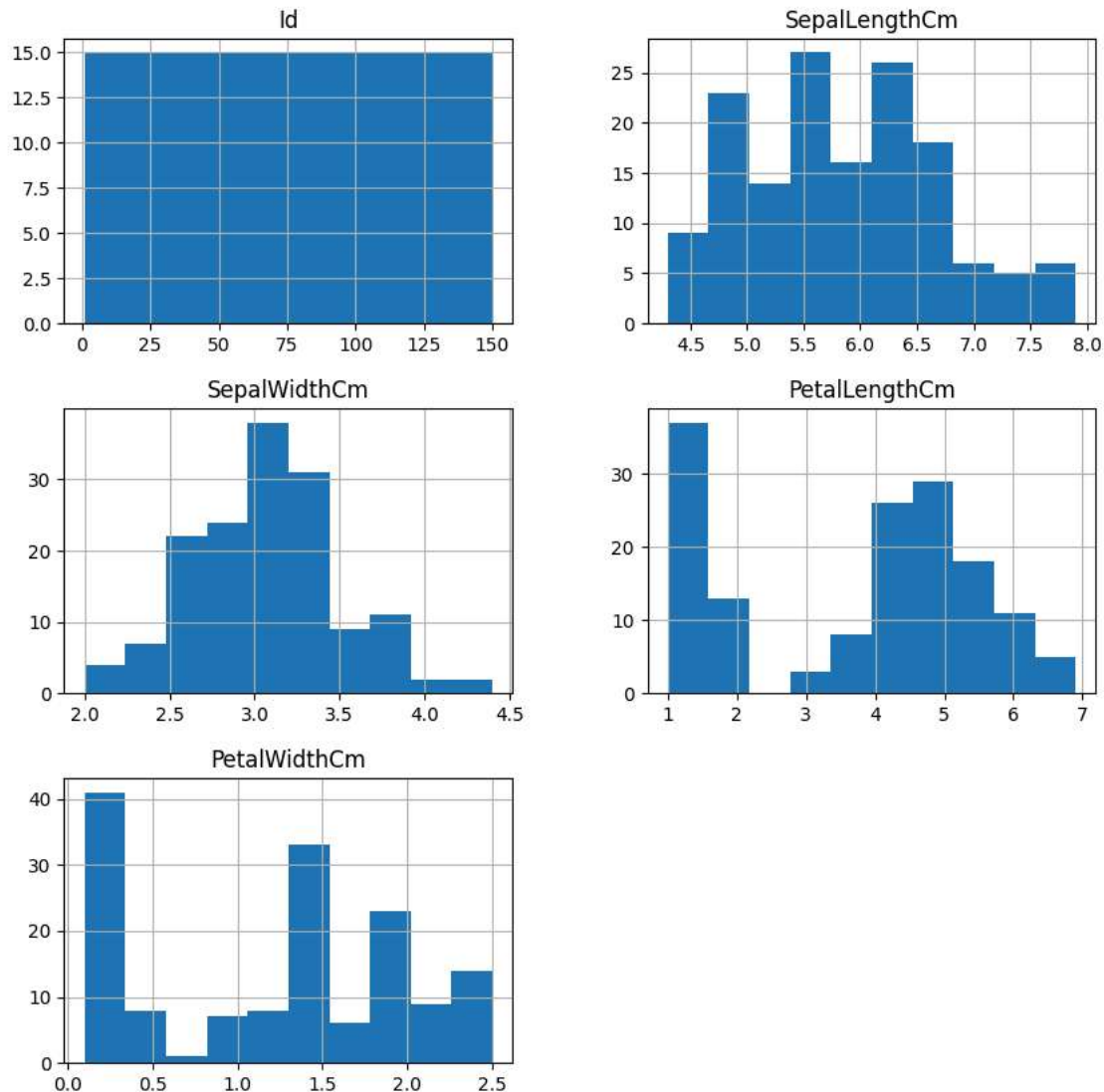
```
print('Number of samples in each class:',df['Species'].value_counts())
```

```
Number of samples in each class: Species
Iris-setosa      50
Iris-versicolor  50
Iris-virginica   50
Name: count, dtype: int64
```

Visualize distribution of individual features

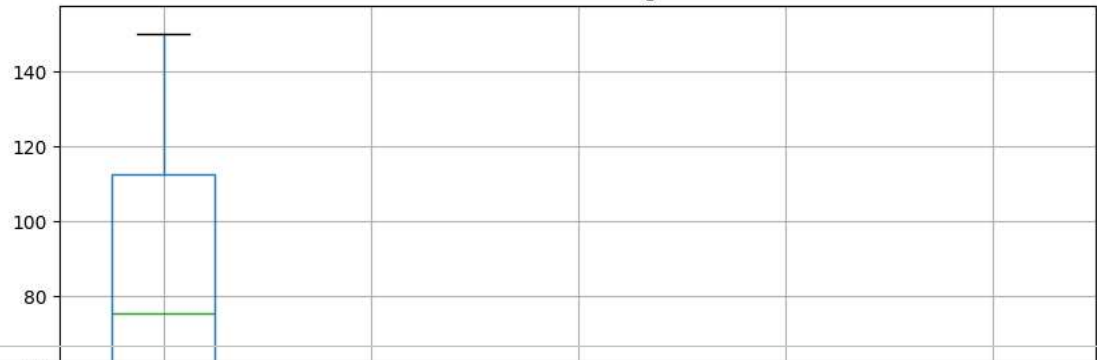
```
import matplotlib.pyplot as plt
```

```
df.hist(figsize=(10,10))
plt.show()
```

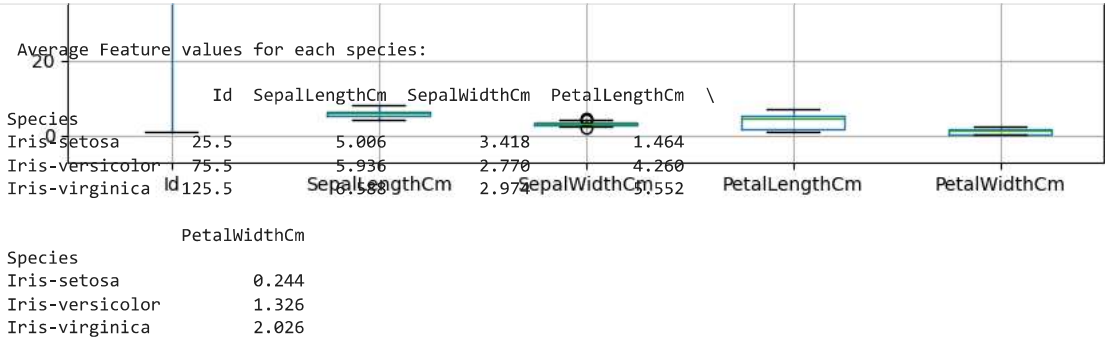


```
df.boxplot(figsize=(10,6))
plt.title('Outlier Detection using Box Plot')
plt.show()
```

Outlier Detection using Box Plot



```
print("\n Average Feature values for each species:\n")
species_mean=df.groupby('Species').mean(numeric_only=True)
print(species_mean)
```



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